5. Substances in the Surroundings – Their States and Properties

1. In the paragraph below write 'solid', 'liquid' or 'gas' in each of the blank (brackets) depending on the substance referred to just before.

On a bright sunny day, Riya and Gargi are playing with a ball (.....) in the park. Gargi feels thirsty. So, Riya brings tender coconut water (.....) for her. At the same time, a strong breeze (.....) starts blowing and it also begins to rain (.....). They run back into the house (.....), change their clothes (.....) and then their mother gives them a cup (.....) of hot milk (.....) to drink.

Answer:
solid, liquid, gas, liquid, solid, solid, solid, liquid.

2. Discuss.

(a) Riya pours some water from her bottle into another bottle. Does it change the shape of the water?
Answer:
Yes, the shape of water changes as water is in liquid state. Liquids do not have a shape of its own. They take the shape of the container.

(b) Halima picks up a small stone from the ground and puts it in the water in a dish. Does the shape of the stone change?
Answer:
No, the shape of the stone does not change. Stone is a solid, hence retains its shape.

3. Write the properties of these substances.

(water, glass, chalk, iron ball, sugar, salt, flour, coal, soil, pen, ink, soap)

Answer:

Substance	State	Properties
1. Water	Liquid	Fluidity, density, solubility, transparency, thermal conductivity.
2. Glass	Solid	Brittleness, hardness, density, transparency.
3. Chalk	Solid	Brittleness, density.
4. Iron ball	Solid	Hardness, density, malleability, ductility, electrical ductility, conductivity, thermal conductivity, luster, sonority.
5. Sugar	Solid	Brittleness, density, solubility.
6. Salt	Solid	Brittleness, density, solubility.
7. Flour	Solid	Density, solubility.
8. Coal	Solid	Brittleness, density, thermal conductivity.
9. Soil	Solid	Brittleness, density.
10. Pen	Solid	Hardness, density.
11. Ink	Liquid	Fluidity, density, solubility.
12. Soap	Solid	Brittleness, hardness, density, solubility.

4. What is sublimation? Write the names of everyday substances that sublimate.

Answer:

1. The change of a solid substance directly into a gas or vapour without first changing into liquid is called sublimation.
2. Substances like: Camphor, napthalene balls, ammonium chloride, iodine.

5. What is made from? Why?

a. A sickle to cut sugarcane.
b. The sheets used for roofing.
c. A screwdriver
d. A pair of tongs.
e. Electric cables.
f. Ornaments.
g. Pots and pans.

(a) A sickle to cut sugarcane.

Answer:

A sickle is made of iron. An iron sickle is hard and malleable. When it is sharpened it will be able to cut the hard sugarcane.

(b) The sheets used for roofing:

Answer:

1. The sheets used for roofing are made of plastic, aluminium.
2. Plastic is hard, hence, protects against weather conditions.
3. Plastic is transparent, thus sunlight can pass through it.

4. Aluminium is hard, light weight and durable, hence, protects against all weather conditions.
5. Malleable hence formed into thin sheets.

(c) A screwdriver:

Answer:

1. A screwdriver is made up of iron, steel, aluminium.
2. A screwdriver possesses property of hardness hence, it easily pierces a screw in piece of wood, wall, metals etc.

(d) A pair of tongs:

Answer:

1. A pair of tongs are made up of iron, steel aluminium etc. Tongs are used to lift hot, boiling utensils or vessels.
2. Tongs are hard, ductile and malleable.
3. Hence, have strong grip to hold utensils.
4. Rubbers fitted on the ends will protect from thermal conduction, from bums.

(e) Electric cables:

Answer:

1. Electric cables are metal wires (thin) wound in plastic.
2. Metal wires possess the property of hardness, ductility, electrical conductivity.
3. Plastic /rubber covering possesses the property of hardness, elasticity and are bad conductors of heat and electricity.

(f) Ornaments:
Answer:

1. They are made up of metals like gold and silver.
2. They possess the property of hardness, ductility, malleability, lustre.

(g) Pots and pans:

Answer:

1. They are used to cook food, hence metals like aluminium, steel are used.
2. They possess the property of hardness, ductility, malleability, thermal conductivity, electrical conductivity, (microwave ovens)

6. What will happen if ….? And why?

(a) Nails are made of plastic
Answer:
If nails are made of plastic, they will not be able to pierce through other substances on being pushed or forced by a hammer. Plastic lacks the property of hardness.

(b) A bell is made of wood.
Answer:

1. If a bell is made of wood it will never make a ringing sound. A wooden bell does not have the property of being sonorous.
2. Sonority is the property of metals to produce a ringing sound.

(c) Rubber is not fitted on a pair of tongs.
Answer:

1. Rubber is a bad conductor of heat and electricity. It will not allow heat to pass to the hands/handle of the tongs, thus protecting us.
2. Pair of tongs are made up of metals which conduct heat and electricity. They have file property of thermal conduction and electrical conduction.
3. If rubber is not fitted on a pair of tongs, we will not be able to lift hot objects with it.

(d) A knife is made of wood.
Answer:
Wood does not have the property of malleability. Therefore, the edge of wooden knife will be blunt. Hence, we will not be able to cut anything with it.

(e) An axe is made of rubber.
Answer:

1. If an axe is made of rubber, it will not be used to cut wood or tree.
2. Rubber does not have the property of hardness that is required to push through to cut it.

7. Who am I?

(a) I'm found in a thermometer, I measure your temperature.
Answer:
Mercury

(b) I make things hot or cold.
Answer:
Heat

(c) I have no shape whatsoever!
Answer:
Liquid, gases

(d) I dissolve in water, but not in kerosene.
Answer:
Salt

8. Why does this happen?

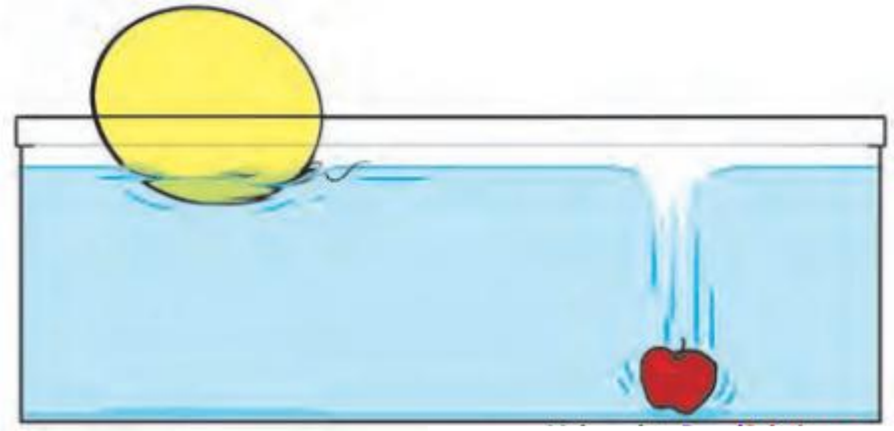
(a) Coconut oil thickens in winter.
Answer:
Coconut oil is in liquid state. In winter the surrounding temperature / atmospheric temperature starts decreasing. Coconut oil starts cooling or losing heat, it changes to solid state.
Thus coconut oil thickens in winter.

(b) Kerosene left open in a dish disappears.
Answer:
When kerosene is left open in a dish, it is exposed to surrounding temperature. As the temperature is more, kerosene starts continuously evaporating and finally disappears.

(c) The fragrance of incense sticks lighted in one corner of a room spreads to the other corner.
Answer:
The fragrance of incense sticks is given out in the form of scented vapours. As vapours are in gaseous state, the gas molecules spread

out in the room. The molecules of gas move very fast and there are no forces to stop them from going apart. Therefore the fragrance of incense sticks lighted in one corner of room spreads to the other corner.

(d) What you see in the picture.



Answer:
The mass of plastic ball is less than an apple. This difference is because of their densities. Since an apple has greater density, it will sink to the bottom on other hand the plastic ball has lesser density, it will float over water surface.